



# **IoT-Based Smart Aeroponic System for Indoor Cultivation of Saffron (Crocus sativus L.)**

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## **ABSTRACT**

Saffron (*Crocus sativus* L.) is one of the most valuable spice crops and demands controlled environmental conditions for optimal yield. This work presents a hardware-implemented Internet of Things (IoT) based aeroponic system for indoor saffron cultivation. The system employs sensors and actuators managed by an ESP32 microcontroller to maintain ideal temperature, humidity, and light levels. Using DHT22 and BH1750 sensors, a Peltier-based cooling unit, ultrasonic mist maker, and RGB LEDs, the system ensures stable environmental parameters. Real-time monitoring and control were achieved via a web dashboard. Experiments maintained temperatures between 18–25°C, humidity between 70–80%, and light intensity near 500 lux, leading to improved stigma yield. This work demonstrates a reliable IoT-enabled approach for high-value saffron cultivation.

**Keywords**— Aeroponics, IoT, Saffron, Smart Farming, Automation, Web Dashboard

**Keywords**: IIoT, Blockchain, Artificial Intelligence, Predictive Monitoring, Cold Chain, Healthcare, On-Device AI, Audit Trails

## **1. Introduction**

Saffron (*Crocus sativus* L.) is the world's most expensive spice, prized for its medicinal, aromatic, and colouring properties. However, saffron cultivation is geographically limited to specific climates such as those in Iran and Kashmir. Traditional soil-based methods face challenges like labour intensity, environmental dependency, and disease [1].susceptibility. Aeroponic cultivation provides a soil-free environment where nutrients are delivered as a fine mist, enabling better oxygenation and control.

By integrating aeroponics with IoT technology, the farming process becomes fully automated. IoT enables real-time monitoring and remote control, reducing manual labour while optimizing environmental parameters. This research develops and evaluates an IoT-controlled aeroponic prototype that maintains ideal growth conditions for saffron through sensors, actuators, and a custom-built web dashboard.

## **2. RESEARCH METHODOLOGY**

The proposed hardware prototype consists of three modules: sensing, control, and cloud communication. An ESP32 microcontroller serves as the central processing unit. Temperature and humidity are sensed using DHT22, while a BH1750 module measures light intensity. The actuators include a Peltier cooling unit with a fan, an ultrasonic mist maker, and RGB LED grow lights. The collected data is transmitted to a web-based IoT dashboard for visualization and control through Wi-Fi.

## 2. Methodology

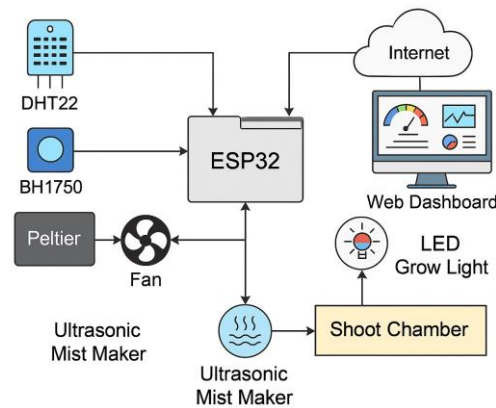


Fig.1 IoT-Based Smart Aeroponic System for Indoor *Crocus sativus L.*

### System Description: IoT-Based Smart Aeroponic System

The proposed system employs an ESP32 microcontroller as the central control and communication unit for automating and monitoring the growth environment of *Crocus sativus L.* (Saffron) in an indoor aeroponic setup.

The block diagram of the developed system is shown in Fig. 1.

#### 1. ESP32 Microcontroller

The ESP32 serves as the core of the system, integrating data acquisition, control logic, and communication functions.

It has built-in Wi-Fi and Bluetooth capabilities, enabling seamless connectivity with cloud servers and remote dashboards.

All sensors and actuators are interfaced to the ESP32 to maintain optimal environmental conditions.

#### 2. Sensors

- **DHT22 Sensor:**  
Measures temperature and humidity inside the shoot chamber. The collected data are used by the ESP32 to regulate the cooling system and mist generation cycles.
- **BH1750 Light Sensor:**  
Monitors the ambient light intensity. The sensor helps the controller maintain ideal light levels by switching the LED Grow Light ON or OFF automatically.

#### 3. Actuators

- **Ultrasonic Mist Maker:**  
Generates fine nutrient-rich mist droplets that supply water and nutrients directly to the plant roots in an aeroponic manner. The ESP32 controls the operation based on humidity and timing parameters to prevent waterlogging or dryness.
- **Peltier Cooling Module and Fan:**  
The Peltier module maintains the desired temperature inside the chamber. The fan assists in uniform air circulation and heat dissipation. Both are controlled by the ESP32 using relay drivers depending on DHT22 feedback.
- **LED Grow Light:**  
Provides artificial light to support photosynthesis. The intensity and ON/OFF timing are controlled automatically according to the BH1750 sensor readings or via remote user commands.

#### 4. Cloud Connectivity and Dashboard

The ESP32 transmits all environmental parameters—temperature, humidity, and light intensity—to the IoT cloud server using its Wi-Fi module.

The data are visualized in real time on a web dashboard, allowing the user to:

- Monitor live environmental data
- Analyze performance trends
- Control actuators (fan, mist maker, lights) remotely

This cloud-based monitoring enables precision farming and data-driven optimization of plant growth conditions.

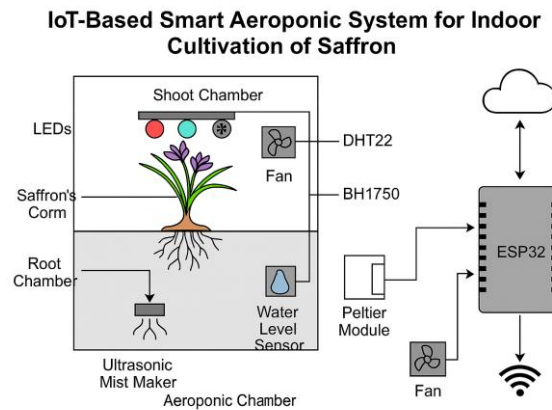
#### 5. Shoot Chamber

The shoot chamber is the enclosed growth environment for the *Crocus sativus L.* plantlets.

It contains:

- LED grow lights
- Ultrasonic mist maker
- Sensors for feedback monitoring

This ensures closed-loop control, maintaining ideal conditions for aeroponic growth — consistent humidity, stable temperature, and controlled light exposure



**Fig. 2. Aeroponic chamber layout showing root, shoot, and sensor placements**

### System Architecture: IoT-Based Smart Aeroponic System for Indoor Cultivation of Saffron

The proposed system employs an *ESP32 microcontroller* as the central processing and control unit for managing environmental conditions required for indoor aeroponic cultivation of *Crocus sativus L.* (saffron).

The complete system architecture is shown in Fig. 2, consisting of multiple sensors, actuators, and communication modules integrated to form a smart, self-regulating growth chamber.

### 2. ESP32 Microcontroller Unit

The ESP32 acts as the main controller and IoT node.

It collects data from sensors (temperature, humidity, light intensity, and water level), processes it, and automatically controls actuators (fan, mist maker, LEDs, and Peltier module).

Its inbuilt Wi-Fi module enables real-time data transmission to a cloud platform, where users can view environmental parameters and control actions remotely through a web or mobile dashboard.

### 3. Sensors

- **DHT22 Sensor:**  
Placed inside the shoot chamber to measure temperature and relative humidity.  
The ESP32 adjusts cooling or misting mechanisms based on these values to maintain optimal growth conditions.
- **BH1750 Light Sensor:**  
Measures light intensity in the shoot chamber.  
If the light level drops below a threshold, the ESP32 turns ON the LED grow lights to maintain the necessary illumination for photosynthesis.
- **Water Level Sensor:**  
Installed in the aeroponic nutrient reservoir to monitor water levels.  
When the level falls below a safe limit, it triggers a notification or can automatically refill the reservoir if connected to a solenoid valve.

### 4. Actuators

- **Ultrasonic Mist Maker:**  
Converts nutrient solution into fine mist droplets that nourish the saffron roots directly without soil.  
The mist maker is controlled by the ESP32 based on humidity and timing cycles.
- **Peltier Module and Fan:**  
Together form a cooling unit that regulates the internal temperature of the chamber.  
The Peltier module absorbs excess heat, and the fan assists in air circulation and heat dissipation.
- **LED Grow Lights:**  
Provide artificial light for saffron's shoot development, controlled automatically based on BH1750 sensor readings.

### 5. IoT and Cloud Integration

The ESP32 transmits real-time environmental data to a cloud server using Wi-Fi.

A web dashboard enables users to:

- Monitor live readings (temperature, humidity, light intensity, and water level)
- Visualize trends and data logs
- Control devices (LED, fan, mist maker) remotely

This connectivity ensures remote supervision, reduced labour, and efficient data-driven cultivation.

### 3. RESULTS AND DISCUSSION

The system was tested for 30 days in a controlled indoor environment. Data from the web dashboard confirmed stable control of temperature (18–25°C), humidity (70–80%), and illumination (450–500 lux). The LED light cycle followed a 12-hour ON/OFF pattern, promoting corm sprouting and stigma elongation. Real-time monitoring graphs displayed consistent environmental stability.

| Parameter                 | Manual System | IoT-Controlled System |
|---------------------------|---------------|-----------------------|
| Temperature (°C)          | 22–30         | 18–25                 |
| Humidity (%)              | 55–65         | 70–80                 |
| Light Intensity (lux)     | Natural ~250  | LED ~500              |
| Flowering Rate (%)        | 60            | 85                    |
| Average Stigma Yield (mg) | 9.2           | 11.5                  |

**Table 1 compares manual versus IoT-based cultivation outcomes.**

#### 1. Temperature Control

- In the manual system, the temperature fluctuated between 22–30°C, depending on ambient conditions, leading to inconsistent growth.
- In the IoT-controlled system, the temperature was maintained within the optimal range of 18–25°C through automatic regulation using the Peltier module and fan, providing a stable environment for saffron corms and shoots.

#### 2. Humidity Regulation

- Manual cultivation maintained humidity between 55–65%, often insufficient for aeroponic mist absorption.
- The IoT-based system achieved higher humidity levels (70–80%) by automatically controlling the ultrasonic mist maker based on feedback from the DHT22 sensor. This promoted better root hydration and nutrient uptake.

#### 3. Light Intensity

- The manual setup relied on natural sunlight (~250 lux), which varied throughout the day.
- The IoT-controlled setup used LED grow lights with adjustable intensity (~500 lux) managed by the BH1750 light sensor, ensuring consistent illumination and improved photosynthesis in the shoot chamber.

#### 4. Flowering Rate

- The flowering rate improved from 60% (manual) to 85% (IoT-controlled). This increase is due to precise control of temperature, humidity, and light, which collectively created ideal conditions for floral initiation and growth.

#### 5. Stigma Yield

- The average stigma yield (the key commercial product from saffron) increased from 9.2 mg to 11.5 mg per flower in the IoT-controlled system. This improvement demonstrates how automation and environmental optimization significantly enhance productivity and crop quality.

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